

Submission PY1-MID-SUB06

Question PY1-MID-Q01

notation changes halfway:

1. range(4) gives 0 through 3
2. keep even i: 0, 2
3. squares $\approx$ $> [0, 4]$

marks, subscripts, and units are mixed up

result written as: [0, 4] (please interpret)

Question PY1-MID-Q02

notation changes halfway:

1. counts $\approx$ Counter(xs)
2. for x in xs: if counts[x] $\approx$ 1: return x
3. return None; two linear passes $\Rightarrow O(n)$ time, $O(n)$ space

marks, subscripts, and units are mixed up

result written as: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (please interpret)

Question PY1-MID-Q03

notation changes halfway:

1. A single pass fixes only adjacent inversions encountered once.
2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.

marks, subscripts, and units are mixed up

result written as: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (please interpret)

Question PY1-MID-Q04

notation changes halfway:

1. linear scan: up to 64 comparisons, $\Theta(n)$
2. binary search: at most $\lceil \log_2(64) \rceil + 1$ comparisons, $\Theta(\log n)$

marks, subscripts, and units are mixed up

result written as: linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7 (please interpret)

Question PY1-MID-Q05

notation changes halfway:

1. Removing shifts later elements left while the iterator advances.
2. Build a filtered list or iterate over a copy, then replace the original.

marks, subscripts, and units are mixed up

result written as: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (please interpret)

Question PY1-MID-Q06

notation changes halfway:

1. range(4) gives 0 through 3

2. keep even i: 0, 2

3. squares $\approx$ $>$ [0, 4]

marks, subscripts, and units are mixed up

result written as: [0, 4] (please interpret)

Question PY1-MID-Q07

notation changes halfway:

1. counts $\approx$ Counter(xs)

2. for x in xs: if counts[x] $\approx$ 1: return x

3. return None; two linear passes $\Rightarrow$ $O(n)$ time, $O(n)$ space

marks, subscripts, and units are mixed up

result written as: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (please interpret)

Question PY1-MID-Q08

notation changes halfway:

1. A single pass fixes only adjacent inversions encountered once.

2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.

marks, subscripts, and units are mixed up

result written as: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (please interpret)